



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: Jetty 1	AREA: B
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Cantilever
SCAFFOLD DRAWING No: PMS-WA-ARCO-039	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 - Wind Actions
- BS EN 1993-1-1:2005 - Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 - Temporary Works Equipment - Scaffolds

SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a Cantilever scaffold required for maintenance of the Fender at Jetty 1 / B. The scaffold envelope is 6.8m x 2.2m x 7.3m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, access, and support/tie arrangements as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Note: This is a category 1 engineered scaffold. Scaffold shall be inspected and approved for use by assurance engineer before tagging for use.

Role	ID	Name	Signature	Date
DESIGNED BY	PMS/C3-1141	Nnamani, Ifeanyi		
VERIFIED BY	PMS/612C	Olowo, Peter		
CHECKED BY	PMS/6	Omikunle, Akinwale		
REVIEWED BY				
APPROVED BY				

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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Cantilever	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-039	Prepared By: Nnamani, Ifeanyi	

LOAD SUMMARY

TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Handrail (kN)	TOTAL (kN)
X	11.42	2.16	13.58
Z	12.35	1.92	14.27

TOTAL VERTICAL LOAD SUMMARY

Load Case	Load Description	TOTAL LIVE LOAD, FY (kN)
LC 2	LL (Live Load)	8.64
LL Total		8.64

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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor $\gamma_M = 1.1$ (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 23.588$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

LOAD CASES

Load diagrams are shown in [Appendix B](#).

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading was applied to the scaffold working platforms as member UDLs in the STAAD model. The platform load intensity is multiplied by the tributary width to obtain the line load applied to platform supporting members. (LC 2).

Member	Tributary Width (m)	Service Load (kN/m ²)	Member Load (kN/m)
Members 324, 327, 331, 332, 353, 356, 427, 428 (Edge)	0.800 / 2 = 0.400	1.500	0.600
Members 334, 397, 430, 432 (Interior)	(0.400 + 0.400) = 0.800	1.500	1.200

Member Load = Service Load x Tributary Width. Edge members take half the adjacent bay width; interior members take half the bay width from each side.

Governing platform live load intensity used for SWL = **1.5 kN/m²**.

Total applied platform live load = **8.64 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **2.16 kN** (member UDL 0.15 kN/m)

Handrail Load (HLZ) = **1.92 kN** (member UDL 0.15 kN/m)

Wind Load

EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.064134 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **11.42 kN**

Total Wind Load - Z Axis (WLZ) = **12.35 kN**

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Dredger P. - How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m
	z - height of structure under consideration	7.3	m
EN 1991-1-4, Sec 4.3	C _{r0} (z) - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	C _o (z) - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor K _r = 0.19 x [z0/z0,ii] ^{0.07}	0.156	
EN 1991-1-4, Table 4.1	Z _{min} < z -> Case 1 applies		
	Wind Turbulence		
	C _r (z) = K _r x ln(z/z0)	1.2166	
	Mean velocity V _m (z) = C _r (z) x C _o (z) x V _b	30.543	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity I _v (z) = k1 / [C _o (z) x ln(z/z0)]	0.1283	
	Peak Velocity Pressure		
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	q _p (z) = [1 + 7*I _v (z)] x 0.5 x rho x V _m (z)²	1106.5235	N/m²
Wind Load = C_f x q_p x A_f			
C _f =		1.2	
q _p =		1.106523	kN/m²
A _f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.064134	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z |
CW = Counterweight (LC 7, LOADTYPE Dead; $\gamma_{stb} = 0.9$ in ULS | 1.0 in SLS)

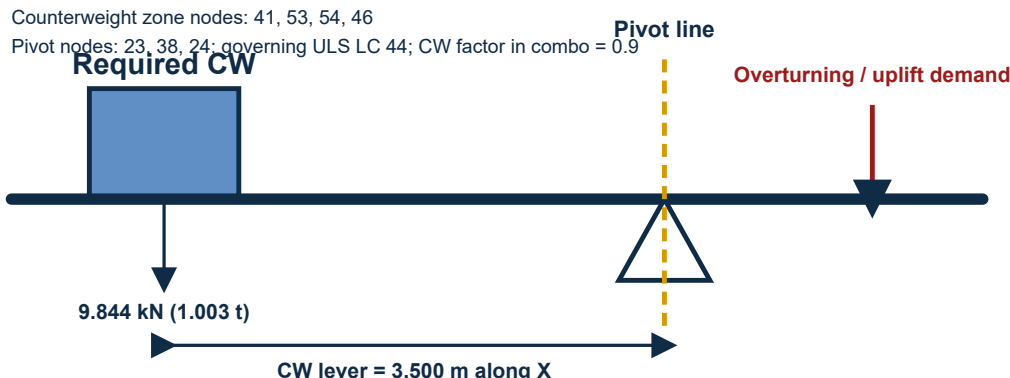
LC No.	Combination	Component Loadings & Factors
8	SLS DL + CW	LC1 x 1.0 + LC7 x 1.0
9	SLS DL + LL + CW	LC1 x 1.0 + LC2 x 1.0 + LC7 x 1.0
10	SLS DL + HLX + CW	LC1 x 1.0 + LC3 x 1.0 + LC7 x 1.0
11	SLS DL - HLX + CW	LC1 x 1.0 + LC3 x -1.0 + LC7 x 1.0
12	SLS DL + HLZ + CW	LC1 x 1.0 + LC4 x 1.0 + LC7 x 1.0
13	SLS DL - HLZ + CW	LC1 x 1.0 + LC4 x -1.0 + LC7 x 1.0
14	SLS DL + LL + HLX + CW	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC7 x 1.0
15	SLS DL + LL - HLX + CW	LC1 x 1.0 + LC2 x 1.0 + LC3 x -1.0 + LC7 x 1.0
16	SLS DL + LL + HLZ + CW	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC7 x 1.0
17	SLS DL + LL - HLZ + CW	LC1 x 1.0 + LC2 x 1.0 + LC4 x -1.0 + LC7 x 1.0
18	SLS DL + WLX + CW	LC1 x 1.0 + LC5 x 1.0 + LC7 x 1.0
19	SLS DL - WLX + CW	LC1 x 1.0 + LC5 x -1.0 + LC7 x 1.0
20	SLS DL + WLZ + CW	LC1 x 1.0 + LC6 x 1.0 + LC7 x 1.0
21	SLS DL - WLZ + CW	LC1 x 1.0 + LC6 x -1.0 + LC7 x 1.0
22	SLS DL + LL + WLX + CW	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0 + LC7 x 1.0
23	SLS DL + LL - WLX + CW	LC1 x 1.0 + LC2 x 1.0 + LC5 x -1.0 + LC7 x 1.0
24	SLS DL + LL + WLZ + CW	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC7 x 1.0
25	SLS DL + LL - WLZ + CW	LC1 x 1.0 + LC2 x 1.0 + LC6 x -1.0 + LC7 x 1.0
26	ULS 1.5DL + 0.9CW	LC1 x 1.5 + LC7 x 0.9
27	ULS 1.5DL + 1.5LL + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC7 x 0.9
28	ULS 1.5DL + 1.5HLX + 0.9CW	LC1 x 1.5 + LC3 x 1.5 + LC7 x 0.9
29	ULS 1.5DL - 1.5HLX + 0.9CW	LC1 x 1.5 + LC3 x -1.5 + LC7 x 0.9
30	ULS 1.5DL + 1.5HLZ + 0.9CW	LC1 x 1.5 + LC4 x 1.5 + LC7 x 0.9
31	ULS 1.5DL - 1.5HLZ + 0.9CW	LC1 x 1.5 + LC4 x -1.5 + LC7 x 0.9
32	ULS 1.5DL + 1.5LL + 1.5HLX + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC7 x 0.9
33	ULS 1.5DL + 1.5LL - 1.5HLX + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC3 x -1.5 + LC7 x 0.9
34	ULS 1.5DL + 1.5LL + 1.5HLZ + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC7 x 0.9
35	ULS 1.5DL + 1.5LL - 1.5HLZ + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC4 x -1.5 + LC7 x 0.9
36	ULS 1.5DL + 1.5WLX + 0.9CW	LC1 x 1.5 + LC5 x 1.5 + LC7 x 0.9
37	ULS 1.5DL - 1.5WLX + 0.9CW	LC1 x 1.5 + LC5 x -1.5 + LC7 x 0.9
38	ULS 1.5DL + 1.5WLZ + 0.9CW	LC1 x 1.5 + LC6 x 1.5 + LC7 x 0.9
39	ULS 1.5DL - 1.5WLZ + 0.9CW	LC1 x 1.5 + LC6 x -1.5 + LC7 x 0.9
40	ULS DL + 1.5WLX + 0.9CW	LC1 x 1.0 + LC5 x 1.5 + LC7 x 0.9
41	ULS DL - 1.5WLX + 0.9CW	LC1 x 1.0 + LC5 x -1.5 + LC7 x 0.9
42	ULS DL + 1.5WLZ + 0.9CW	LC1 x 1.0 + LC6 x 1.5 + LC7 x 0.9
43	ULS DL - 1.5WLZ + 0.9CW	LC1 x 1.0 + LC6 x -1.5 + LC7 x 0.9
44	ULS 1.5DL + 1.5LL + 1.5WLX + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5 + LC7 x 0.9
45	ULS 1.5DL + 1.5LL - 1.5WLX + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC5 x -1.5 + LC7 x 0.9
46	ULS 1.5DL + 1.5LL + 1.5WLZ + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC7 x 0.9
47	ULS 1.5DL + 1.5LL - 1.5WLZ + 0.9CW	LC1 x 1.5 + LC2 x 1.5 + LC6 x -1.5 + LC7 x 0.9

COUNTERWEIGHT STABILITY CHECK — EQU LIMIT STATE (EN 1990)

Basis: Cantilever analysis-line moment balance about the pivot line per EN 1990 EQU limit state. Negative vertical support reactions from STAAD ULS combinations represent hold-down demand at backspan nodes. ULS action factors (destabilising: 1.5, stabilising CW: 0.9) are already embedded in the STAAD combinations; no additional multiplier is applied here.

Counterweight verified by ULS global moment equilibrium. Maximum ULS hold-down demand at backspan base node 21 = 4.030 kN (LC 46). This is an EQU partial-factor artefact — the overall overturning is resisted by the counterweight moment; no base anchorage to the supporting structure is required provided the counterweight is maintained in position. Ensure counterweight is physically secured and in place before any live loading is applied.

Full design procedure is documented in [Appendix F](#).



Summary Table

Governing ULS Case	Pivot Line / CW Lever	Current CW in STAAD	CW Factor in Combo	Minimum Design CW	Target / Adopted CW
LC 44 ULS 1.5DL + 1.5LL + 1.5WLX + 0.9CW	Nodes 23, 38, 24 3.5 m	10.64 kN 1.085 t	0.9	9.844 kN 1.003 t	9.844 kN 1.003 t

Analysis Line Breakdown

Analysis Line	Governing LC	Critical Nodes	Design Hold-down Moment (kNm)	CW Factor in Combo	Required Design CW (kN)
Z = 0 m	LC 44	21	9.912	0.9	3.147
Z = 0.6 m	LC 44	28	5.56	0.9	1.765
Z = 1.2 m	LC 44	27	5.004	0.9	1.589
Z = 1.8 m	LC 44	26	10.532	0.9	3.343

Counterweight Zone & Load Distribution

Counterweight Zone	Zone Area (m²)	Target Pressure (kN/m²)	Selected Grid	Distributed Load (kN)
Nodes 41, 53, 54, 46	1.8	5.469	Z direction (1.8 m² tributary area)	9.844

Counterweight Supporting Members	Tributary Width (m)	Target Applied UDL (kN/m)
Members 73, 75, 76, 88, 89, 90 (Edge)	0.500 / 2 = 0.250	1.367
Members 81, 82, 83 (Interior)	(0.250 + 0.250) = 0.500	2.734

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.661 at Member No. **293**, Load Combination **47**, Clause **EC-6.2.7(5)**. **PASS** - All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
293	PIP483.6	PASS	EC-6.2.7(5)	0.661	47	6.76	T
31	PIP483.6	PASS	EC-6.3.3-662	0.510	47	17.78	C
30	PIP483.6	PASS	EC-6.3.3-662	0.505	46	17.33	C
572	PIP483.6	PASS	EC-6.3.3-661	0.499	45	15.76	C
253	PIP483.6	PASS	EC-6.2.7(5)	0.477	47	2.95	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
45	PIP483.6	PASS	EC-6.2.9.1	0.422	38	0.50	C
257	PIP483.6	PASS	EC-6.2.9.1	0.397	47	7.07	T
51	PIP483.6	PASS	EC-6.3.3-662	0.395	47	20.76	C
50	PIP483.6	PASS	EC-6.3.3-662	0.363	46	17.86	C
98	PIP483.6	PASS	EC-6.3.3-662	0.360	47	18.71	C
46	PIP483.6	PASS	EC-6.2.9.1	0.333	47	7.14	C
47	PIP483.6	PASS	EC-6.2.9.1	0.332	46	5.78	C
546	PIP483.6	PASS	EC-6.2.9.1	0.322	46	8.37	C
534	PIP483.6	PASS	EC-6.2.9.1	0.317	38	0.40	C
71	PIP483.6	PASS	EC-6.2.9.1	0.281	47	1.50	T
25	PIP483.6	PASS	EC-6.3.3-662	0.270	47	4.89	C
3	PIP483.6	PASS	EC-6.3.3-662	0.252	39	2.61	C
432	PIP483.6	PASS	EC-6.2.9.1	0.248	46	0.07	T
430	PIP483.6	PASS	EC-6.2.9.1	0.242	47	0.07	T
97	PIP483.6	PASS	EC-6.3.3-662	0.240	46	18.18	C
353	PIP483.6	PASS	EC-6.3.3-662	0.237	46	0.14	C
32	PIP483.6	PASS	EC-6.2.9.1	0.233	46	4.57	C
29	PIP483.6	PASS	EC-6.2.9.1	0.232	39	2.94	C
427	PIP483.6	PASS	EC-6.3.3-662	0.229	47	0.11	C
443	PIP483.6	PASS	EC-6.2.9.1	0.229	46	0.02	C
255	PIP483.6	PASS	EC-6.2.9.1	0.224	44	0.70	T
447	PIP483.6	PASS	EC-6.2.9.1	0.219	47	0.07	C
44	PIP483.6	PASS	EC-6.2.9.1	0.213	38	0.65	C
439	PIP483.6	PASS	EC-6.2.9.1	0.205	46	0.88	T
511	PIP483.6	PASS	EC-6.3.3-662	0.205	47	3.66	C
573	PIP483.6	PASS	EC-6.3.3-661	0.201	40	6.18	C
532	PIP483.6	PASS	EC-6.3.3-662	0.193	39	1.93	C
284	PIP483.6	PASS	EC-6.3.3-662	0.186	45	10.69	C
258	PIP483.6	PASS	EC-6.3.3-662	0.184	45	10.59	C
334	PIP483.6	PASS	EC-6.2.9.1	0.184	35	0.42	T
485	PIP483.6	PASS	EC-6.3.3-661	0.183	45	6.83	C
67	PIP483.6	PASS	EC-6.3.3-662	0.182	47	3.74	C
397	PIP483.6	PASS	EC-6.2.9.1	0.182	34	0.41	T
514	PIP483.6	PASS	EC-6.3.3-661	0.182	46	8.28	C
468	PIP483.6	PASS	EC-6.3.3-661	0.178	45	6.65	C
554	PIP483.6	PASS	EC-6.3.3-661	0.177	38	5.12	C
387	PIP483.6	PASS	EC-6.2.9.1	0.174	46	0.20	T
268	PIP483.6	PASS	EC-6.2.9.1	0.173	45	0.00	C
449	PIP483.6	PASS	EC-6.2.9.1	0.173	46	3.23	T
66	PIP483.6	PASS	EC-6.3.3-662	0.172	47	3.22	C
466	PIP483.6	PASS	EC-6.3.3-661	0.172	45	6.38	C
246	PIP483.6	PASS	EC-6.2.9.1	0.169	46	3.28	C
393	PIP483.6	PASS	EC-6.2.9.1	0.168	47	0.18	T
262	PIP483.6	PASS	EC-6.2.9.1	0.166	45	0.05	T
463	PIP483.6	PASS	EC-6.3.3-662	0.166	47	1.20	C

Showing highest 50 utilization ratios; 462 lower-utilization members omitted.

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **8.948 kN** (Member 546, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 8.948 kN <= 10.0 kN -> **PASS**

Class A right-angle couplers are adequate.

Member force table: see [Appendix E](#).

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	10.0	15.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	-	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance x 0.15 / C_F = **11.794 kN**

Coefficient of friction (C_F) = **0.3**

Frictional Resistance (FR) = C_F x Max f_y = **3.5382 kN**

Spring Stiffness = FR / 0.15 = **23.588 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state - values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix E](#).

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 24 Member 555 Member length L = 1800.0 mm	0.298 mm	$L/100 = 1800/100 = 18.0$ mm	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 19 Member 426 Member length L = 2700.0 mm	3.926 mm	$L/200 = 2700/200 = 13.5$ mm	PASS

CONCLUSION

The structural analysis and design of the **Cantilever** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.661** at Member 293). Vertical and horizontal deflections

are within their respective allowable limits of $L/100 = 18.0\text{mm}$ and $L/200 = 13.5\text{mm}$. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

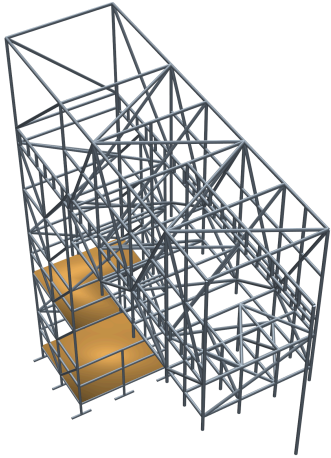
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Cantilever scaffold configuration matches the approved drawing and is suitable for maintenance.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Install and secure cantilever support members, needles, anchors, ties, and back-span restraints exactly as shown on the approved drawing before building the working platform.
8. Build the scaffold frame to the approved envelope of 6.8m x 2.2m x 7.3m, checking level, plumb, and cantilever projection at each lift.
9. Install bracing, ledgers, transoms, platform boards, guardrails, and toe boards progressively, maintaining all specified ties and restraints.
10. Install mid-level horizontal ledgers and transoms at 0.8m height in accordance with the working drawing.
11. Install mid-level horizontal ledgers and transoms at 1.3m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 2.3m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 2.8m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 3.3m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 4.3m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 4.6m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 5.3m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 5.6m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 6.5m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 7.0m height in accordance with the working drawing.
21. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
22. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
23. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
24. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.

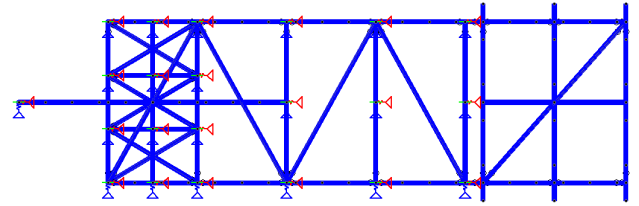
25. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
26. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.
End-user or third-party modification of this scaffold structure is strictly prohibited.
27. For dismantling, perform the erection steps in reverse order.

APPENDIX A - MODEL VIEWS

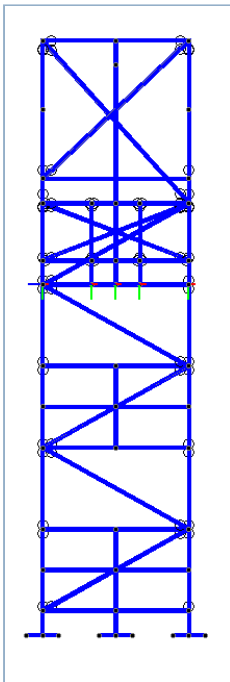
3D Isometric View



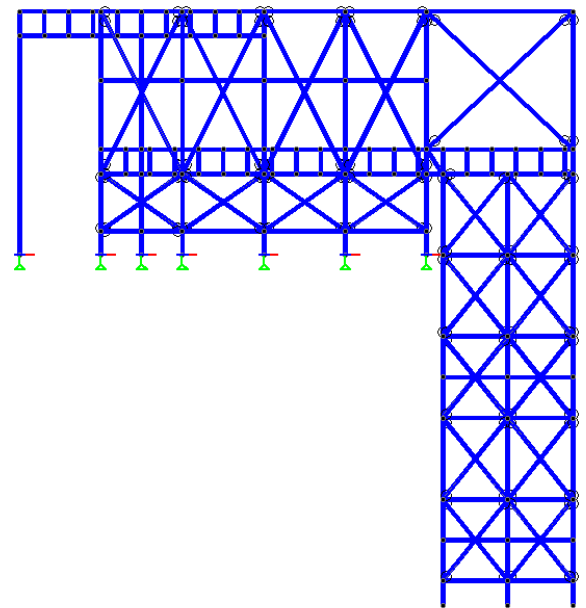
Plan View



Elevation from +X Axis



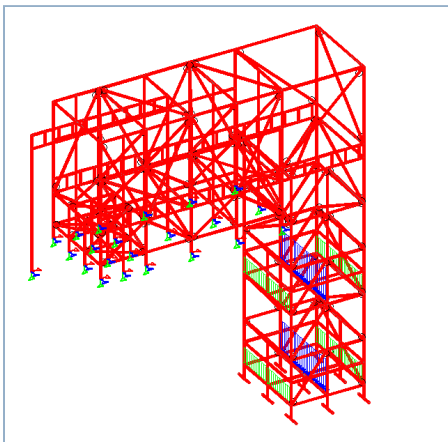
Elevation from +Z Axis



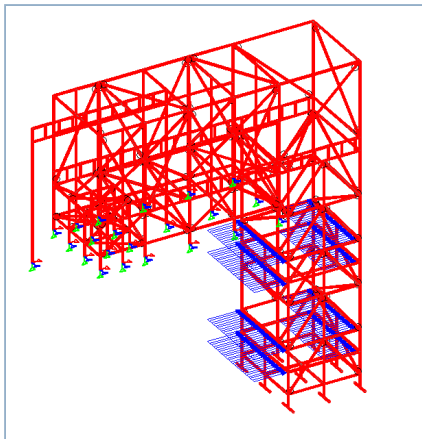
Scaffold Dimensions: W = 6.8m | D = 2.2m | H = 7.3m | **Lifts:** 0.3m, 0.8m, 1.3m, 2.3m, 2.8m, 3.3m, 4.3m, 4.6m, 5.3m, 5.6m, 6.45m, 7.0m, 7.3m

APPENDIX B - LOAD DIAGRAMS

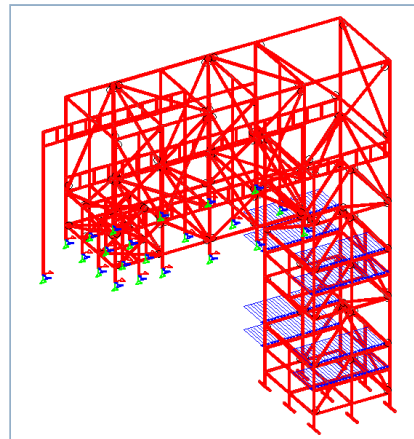
Live Load - Platform Live Load (LL)



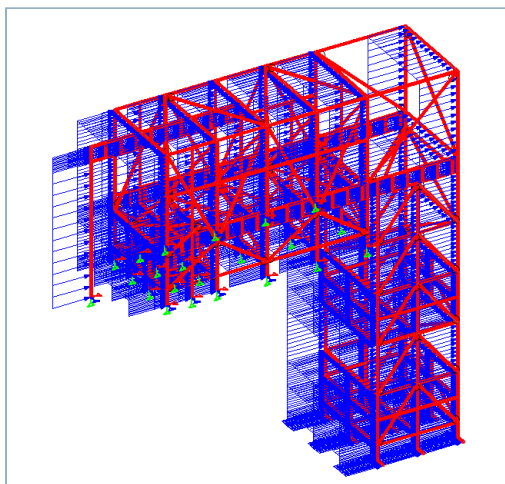
Handrail Load in X Direction (HLX)



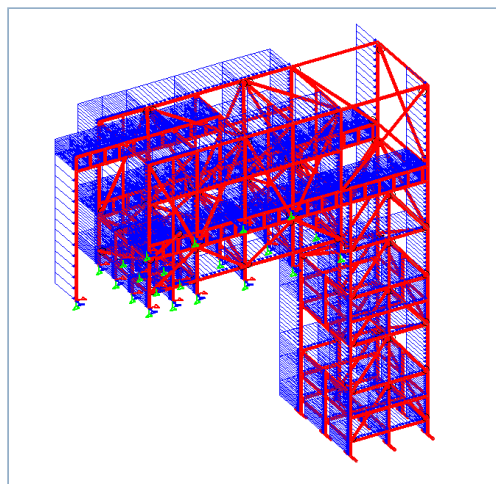
Handrail Load in Z Direction (HLZ)



Wind Load - X Axis (WLX)

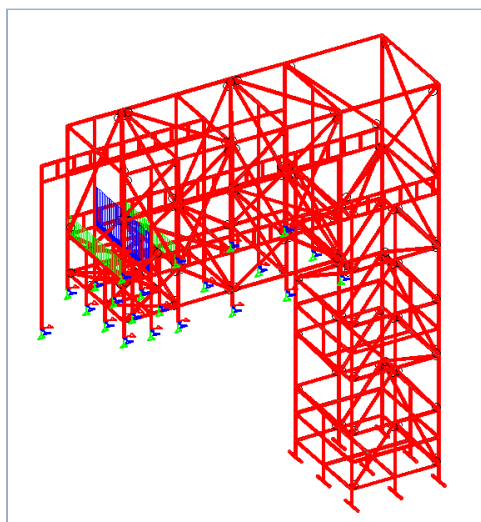


Wind Load - Z Axis (WLZ)



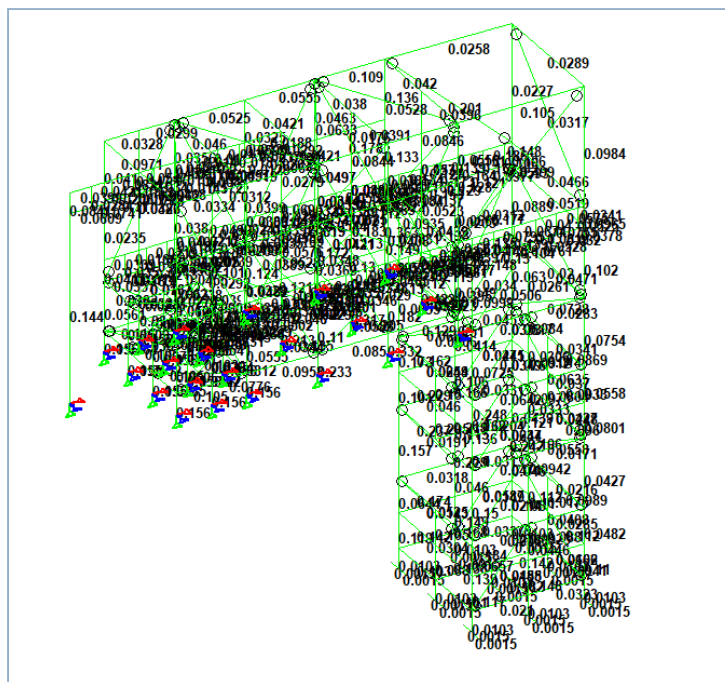
Counterweight Load (CW) — LOADTYPE Dead, LC 7

Counterweight applied as a distributed dead load on the backspan members within the CW zone (nodes 41, 53, 54, 46). Classified as LOADTYPE Dead so that $\gamma_{stb} = 0.9$ is applied by STAAD in ULS combinations.

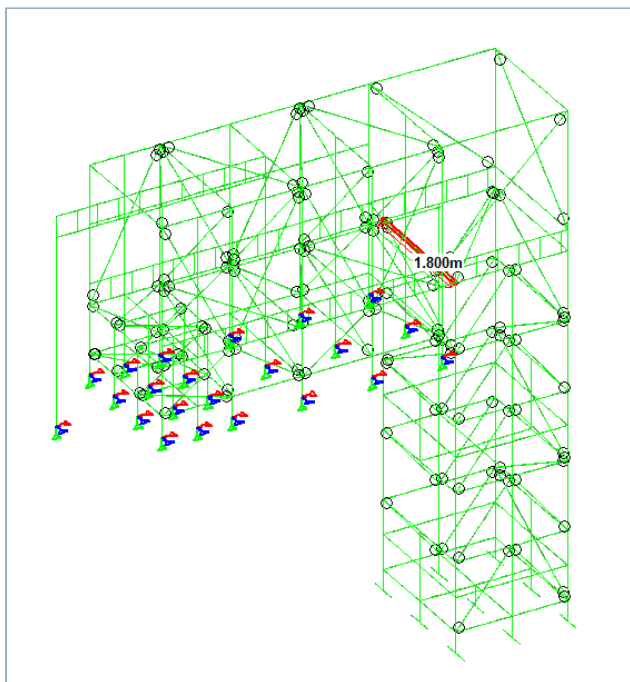


APPENDIX C - ANALYSIS RESULTS

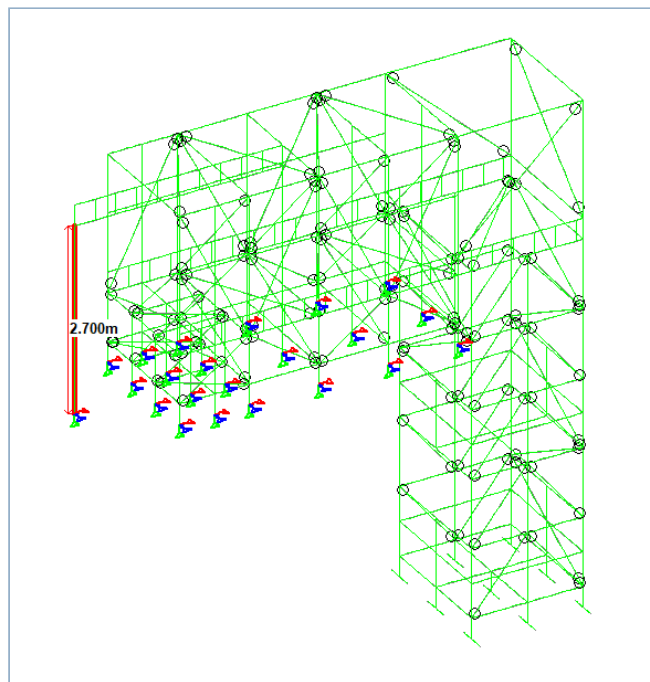
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX E - GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
546	23	43	8.948	0.171	0.080	-0.004	-0.011	0.047
131	23	63	7.826	0.059	-0.038	-0.003	-0.020	-0.029
314	23	176	7.771	0.282	0.019	-0.001	0.010	-0.024
235	23	143	7.753	-0.132	-0.040	-0.001	-0.005	-0.018
241	23	147	7.499	-0.100	-0.028	0.001	0.002	-0.012
247	23	151	7.274	-0.132	-0.032	0.000	0.011	-0.020
284	23	172	7.129	0.613	0.010	-0.000	0.007	0.071
258	23	154	7.063	0.618	0.022	0.001	-0.012	0.110
253	23	151	6.802	0.233	-0.042	0.001	0.005	0.064
131	24	63	6.800	0.021	-0.033	-0.004	-0.006	-0.009
546	19	43	6.614	-0.236	0.071	-0.001	-0.010	-0.002
546	15	43	6.560	0.473	0.005	-0.006	0.001	0.072
314	24	176	6.523	0.045	0.077	-0.000	0.022	-0.021
278	23	168	6.109	0.550	0.011	-0.000	0.003	0.078
116	25	78	6.101	0.003	-0.044	0.002	-0.010	-0.001
284	24	172	6.091	0.410	0.090	-0.003	-0.011	0.044
132	25	65	6.084	0.013	-0.021	0.001	-0.000	-0.001
131	15	63	6.015	0.032	-0.007	-0.003	-0.004	-0.014
235	19	143	6.014	-0.123	-0.040	-0.001	-0.005	-0.017
116	23	78	5.961	-0.036	0.028	0.002	0.013	0.019
264	23	158	5.915	0.627	0.015	-0.000	-0.005	0.097
258	19	154	5.892	0.546	0.019	0.001	-0.011	0.100
314	19	176	5.825	0.289	0.018	-0.001	0.010	-0.018
131	19	63	5.807	0.059	-0.035	-0.002	-0.018	-0.028
241	19	147	5.773	-0.095	-0.028	0.001	0.002	-0.013
132	23	65	5.772	-0.012	0.040	0.001	0.021	0.011
253	23	151	5.762	0.512	0.012	0.002	0.007	0.067

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
555	0.298	0.450	24	6039	6.571	-0.238	65.984
555	0.297	0.450	20	6051	6.584	-0.135	65.991
555	0.291	0.450	21	6185	-6.606	-0.110	-66.004
555	0.290	0.450	25	6207	-6.619	-0.213	-66.011
430	0.284	0.525	24	3164	-6.523	-1.356	79.850
432	0.277	0.375	25	3243	-6.411	-1.408	-80.358
397	0.264	0.375	17	3411	0.514	-1.739	-25.671
334	0.263	0.525	16	3426	0.360	-1.598	24.890
397	0.256	0.375	24	3513	-7.011	-1.430	81.466
334	0.255	0.525	25	3531	-6.826	-1.483	-82.246
397	0.253	0.375	22	3558	24.476	-1.416	-0.493
397	0.251	0.375	14	3581	4.318	-1.503	-0.439
334	0.251	0.525	23	3581	-27.325	-1.787	-0.288
397	0.251	0.375	9	3582	-1.464	-1.619	-0.390
397	0.251	0.375	15	3592	-7.247	-1.734	-0.341
397	0.250	0.375	23	3597	-27.405	-1.820	-0.286
334	0.250	0.525	15	3601	-7.224	-1.680	-0.342
334	0.249	0.525	9	3608	-1.500	-1.544	-0.391
334	0.249	0.525	22	3618	24.325	-1.300	-0.495
334	0.249	0.525	14	3620	4.224	-1.407	-0.440
430	0.246	0.525	16	3663	-2.918	-1.404	24.077
397	0.245	0.375	25	3666	4.082	-1.806	-82.245
334	0.244	0.600	24	3687	2.737	-1.696	81.464
432	0.239	0.375	17	3768	-2.868	-1.393	-24.585
397	0.238	0.300	16	3775	-3.061	-1.615	24.891
334	0.237	0.600	17	3802	-2.970	-1.622	-25.672
430	0.236	0.525	25	4000	24.000	-1.000	-0.000

Max = 0.298 mm (LC 24) | L = Member 555 length (1800 mm) | Allowable L/100 = 18.0 mm -> **PASS**

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
426	3.926	1.350	19	687	-18.442	-0.003	0.007
426	3.912	1.350	22	690	18.538	-0.003	0.025
426	3.912	1.350	23	690	-18.378	-0.003	0.018
426	3.898	1.350	18	692	18.474	-0.003	0.014
426	0.989	1.125	17	2730	0.093	-0.004	7.158
426	0.986	1.125	13	2737	0.022	-0.004	7.147
426	0.981	1.125	12	2753	0.023	-0.004	-7.124
426	0.977	1.125	16	2762	0.093	-0.004	-7.112
426	0.699	1.125	11	3865	-3.458	-0.004	0.009
426	0.684	1.125	14	3946	3.574	-0.003	0.025
426	0.684	1.125	15	3947	-3.387	-0.004	0.021
426	0.670	1.125	10	4032	3.503	-0.004	0.014
110	0.497	1.000	23	4021	-23.895	-0.005	0.018
110	0.480	1.000	19	4164	-23.939	-0.004	0.019
110	0.434	0.833	18	4606	24.009	-0.003	-0.020
531	0.421	0.833	23	4746	-23.860	-0.003	0.004
110	0.417	1.000	22	4795	24.046	-0.004	-0.022
531	0.413	1.000	19	4839	-23.877	-0.003	0.008
531	0.402	1.000	18	4979	23.966	-0.003	-0.005
531	0.394	1.000	22	5072	24.015	-0.003	-0.003
426	0.304	0.450	25	8893	0.133	-0.005	9.282
426	0.302	0.450	21	8949	0.049	-0.006	9.268
426	0.298	0.450	20	9061	0.049	-0.006	-9.242
426	0.296	0.450	24	9125	0.133	-0.005	-9.229
103	0.264	1.000	20	7585	13.835	-0.008	9.169
103	0.262	1.000	24	7620	13.882	-0.006	9.171

Max = 3.926 mm (LC 19) | L = Member 426 length (2700 mm) | Allowable L/200 = 13.5 mm -> **PASS**

APPENDIX F — COUNTERWEIGHT DESIGN CALCULATION (A–Z)

Design approach: An initial counterweight estimate of 1 kN/m² applied uniformly over the counterweight zone was used for the first STAAD analysis run. This is the standard iterative procedure — a non-trivial initial load is necessary to obtain meaningful support reactions, from which the actual counterweight demand is then derived by moment equilibrium (see F.3). The member loads shown in F.5 represent the final design values applied in the current STAAD model.

F.1 — Design Basis

EN 1990:2002 EQU

The scaffold is a cantilever arrangement. Imposed loads on the overhang create an overturning moment about the support edge. Stability is verified at the **Equilibrium (EQU) limit state** per EN 1990:2002 — the appropriate limit state where loss of static equilibrium governs over material failure. The EQU partial factors embedded in the STAAD ULS combinations are:

- Destabilising permanent action (DL): $\gamma_{dst} = 1.5$
- Destabilising variable action (LL, WL): $\gamma_{dst} = 1.5$
- Stabilising permanent action (CW): $\gamma_{stb} = 0.9$

The check is satisfied when the stabilising moment from the counterweight — at its reduced factored value — equals or exceeds the overturning moment from all destabilising actions at their amplified values. This is assessed through the STAAD ULS combination results.

F.2 — Geometry: Pivot Line, Counterweight Zone and Lever Arm

STAAD model geometry

Pivot line: The row of base support nodes at the edge of the supporting structure about which overturning is assessed — nodes **23, 38, 24**, located at $Z = 4.0$ m.

Counterweight zone: The plan polygon on the backspan platform designated for kentledge placement, bounded by nodes **41, 53, 54, 46** at elevation $y = 1.0$ m.

Zone plan area: $A_{CW} = 1.8$ m²

Zone centroid (geometric centre of the polygon) at $Z = 0.5$ m.

Lever arm (perpendicular distance from CW centroid to pivot line):

$$L_{CW} = |0.5 - 4.0| = 3.5 \text{ m}$$

F.3 — Overturning Moment and Minimum Required Counterweight

Moment balance method

Vertical base support reactions (F_Y) are read from the STAAD output for every ULS combination across all base nodes. STAAD sign convention: positive F_Y = upward push from ground (compression); negative F_Y = structure lifting off — a hold-down demand at that node.

For each ULS combination, every node showing negative F_Y contributes to the overturning moment about the pivot line:

$$M_{\text{overturn}} = \sum (|F_{Y,i}| \times d_i)$$

where d_i is the perpendicular distance from node i to the pivot line (m).

The minimum counterweight required to stabilise the moment at that analysis line:

$$CW_{\text{min,line}} = M_{\text{overturn}} \div (Y_{\text{stb}} \times L_{\text{CW}})$$

This is evaluated at each distinct row of base nodes (analysis line) independently for every ULS combination. The line-by-line CW_{min} values within a given combination are summed to obtain the total counterweight demand for that combination — each analysis line represents a separate strip of the scaffold base, and summing across all strips gives the total counterweight required to stabilise the structure as a whole. The combination producing the largest total demand is the governing case; the minimum design counterweight equals that total. Results for the governing combination are tabulated below.

Analysis Line	Governing LC	Critical Node(s)	Σ Overturning Moment, M (kNm)	Y_{stb}	$CW_{\text{min}} = M \div (Y_{\text{stb}} \times 3.5 \text{ m})$
Z = 0 m	LC 44	21	9.912	0.9	$9.912 \div (0.9 \times 3.5) = \mathbf{3.147 \text{ kN}}$
Z = 0.6 m	LC 44	28	5.56	0.9	$5.56 \div (0.9 \times 3.5) = \mathbf{1.765 \text{ kN}}$
Z = 1.2 m	LC 44	27	5.004	0.9	$5.004 \div (0.9 \times 3.5) = \mathbf{1.589 \text{ kN}}$
Z = 1.8 m	LC 44	26	10.532	0.9	$10.532 \div (0.9 \times 3.5) = \mathbf{3.343 \text{ kN}}$
Minimum Design Counterweight = ΣCW_{min} across all lines (LC 44):					9.844 kN (1.003 t)

F.4 — Design Counterweight

Target value

The counterweight is sized at the calculated minimum:

$$CW_{\text{design}} = CW_{\text{min}} = \mathbf{9.844 \text{ kN (1.003 t)}}$$

This is the minimum kentledge mass required within the designated zone to satisfy EQU overturning equilibrium across all ULS combinations. It is used as the basis for STAAD load distribution.

F.5 — Distribution of Counterweight Load to STAAD Members

Load distribution

The counterweight is represented in STAAD as UDL (kN/m) member loads on horizontal members within the CW zone. The equivalent uniform pressure over the zone is first determined:

$$q_{\text{CW}} = CW_{\text{design}} \div A_{\text{CW}} = 9.844 \div 1.8 = \mathbf{5.469 \text{ kN/m}^2}$$

Each horizontal member carries load proportional to its tributary width (edge members: half the adjacent bay width; interior members: sum of adjacent half-bays):

$$w = q_{\text{CW}} \times b_{\text{tributary}}$$

Member(s)	Role	Tributary Width, b	$w = 5.469 \times b \text{ (kN/m)}$
Members 73, 75, 76, 88, 89, 90	Edge	$0.500 / 2 = 0.250 \text{ m}$	$5.469 \times 0.25 = \mathbf{1.367 \text{ kN/m}}$
Total applied to zone members:			9.844 kN

Member(s)	Role	Tributary Width, b	$w = 5.469 \times b$ (kN/m)
Members 81, 82, 83	Interior	$(0.250 + 0.250) = 0.500$ m	$5.469 \times 0.5 = 2.734$ kN/m
Total applied to zone members:			9.844 kN

These UDL GY (downward) loads are applied in STAAD under LC 7, declared as LOADTYPE Dead. This classification causes the EQU stabilising factor of 0.9 to be applied automatically in the ULS combinations, consistent with EN 1990 Table A1.2(A). Refer to [Appendix G](#) for the applicable design standards.

F.6 — Residual Hold-down Demand at Backspan Base Nodes

EQU partial factor effect

Under ULS EQU combinations, the destabilising factor (1.5) exceeds the stabilising CW factor (0.9). As a consequence, backspan base nodes close to the pivot line may continue to register negative F_Y in the STAAD output even with a correctly sized counterweight in place. This is a mathematical consequence of the EQU factor distribution — not an indication of counterweight inadequacy.

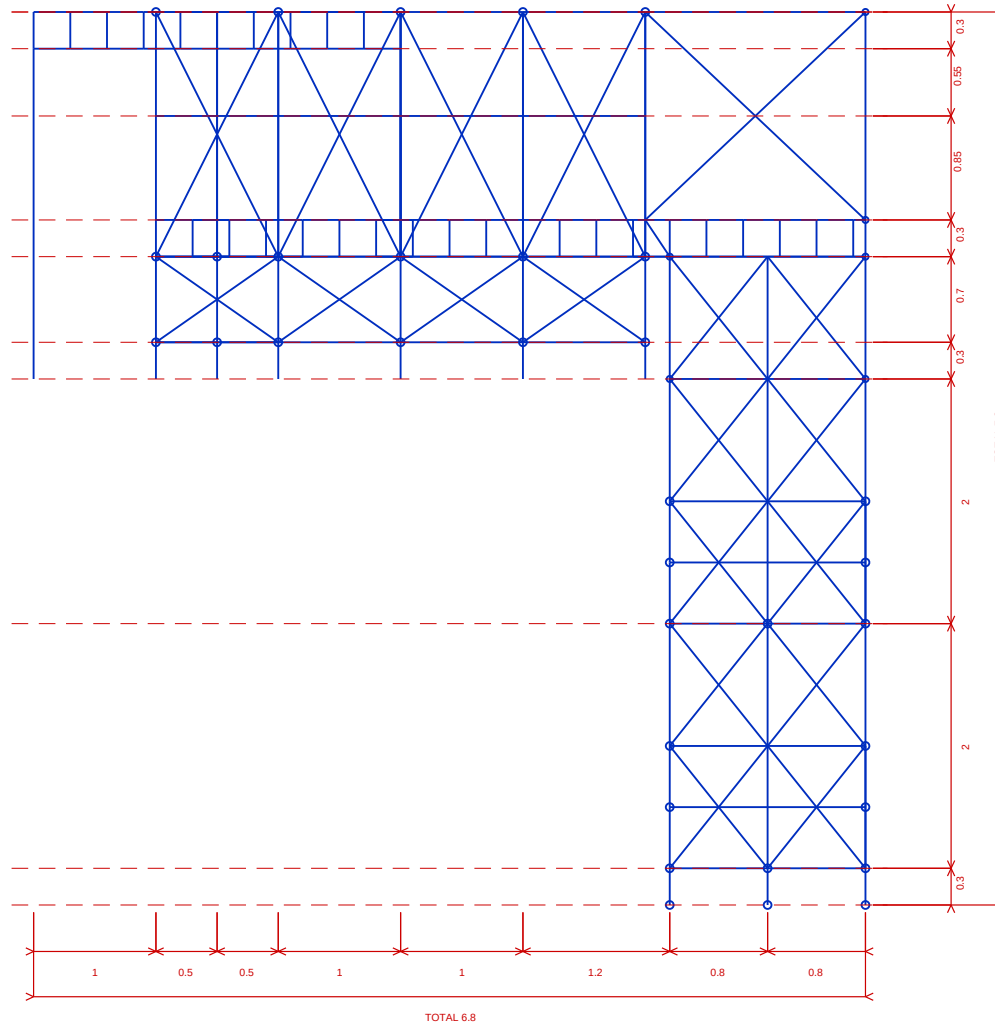
The governing verification criterion is global moment equilibrium about the pivot line, demonstrated in G.3 above. The maximum residual hold-down demand recorded at any single base node is **4.03 kN** at node 21, LC 46 (ULS 1.5DL + 1.5LL + 1.5WLZ + 0.9CW). The counterweight provides the necessary global stabilising moment; no structural anchorage of the scaffold base to the supporting surface is required, provided the counterweight remains in position throughout the working life of the scaffold.

APPENDIX G - REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design (EQU limit state, Table A1.2(A) — Counterweight Stability)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	-
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

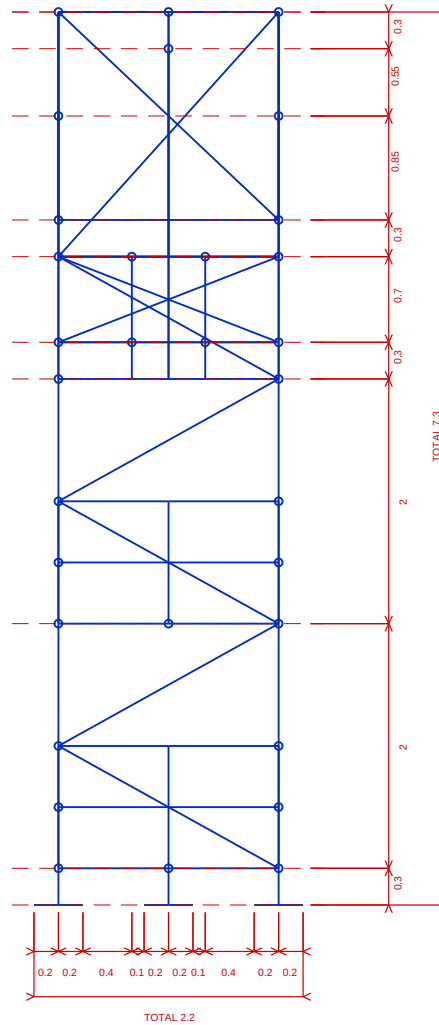
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



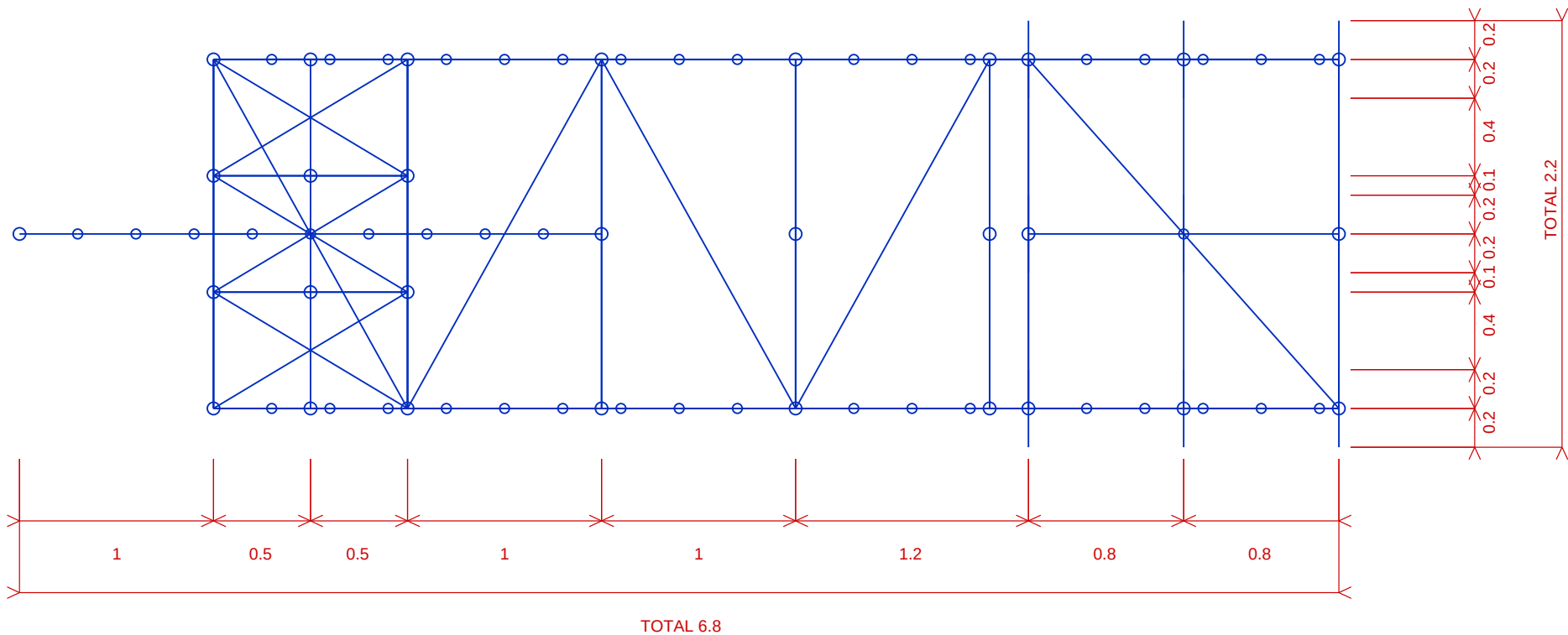
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-040-FRONT-VIEW			 NLNG	
DRAWING TITLE PMS-WA-ARCO-040 - FRONT VIEW		SCALE NTS	DATE 29-JUN-2026	REV 0		
		COMPANY				
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED		
				SHEET 1 OF 4		
		SIZE A4				



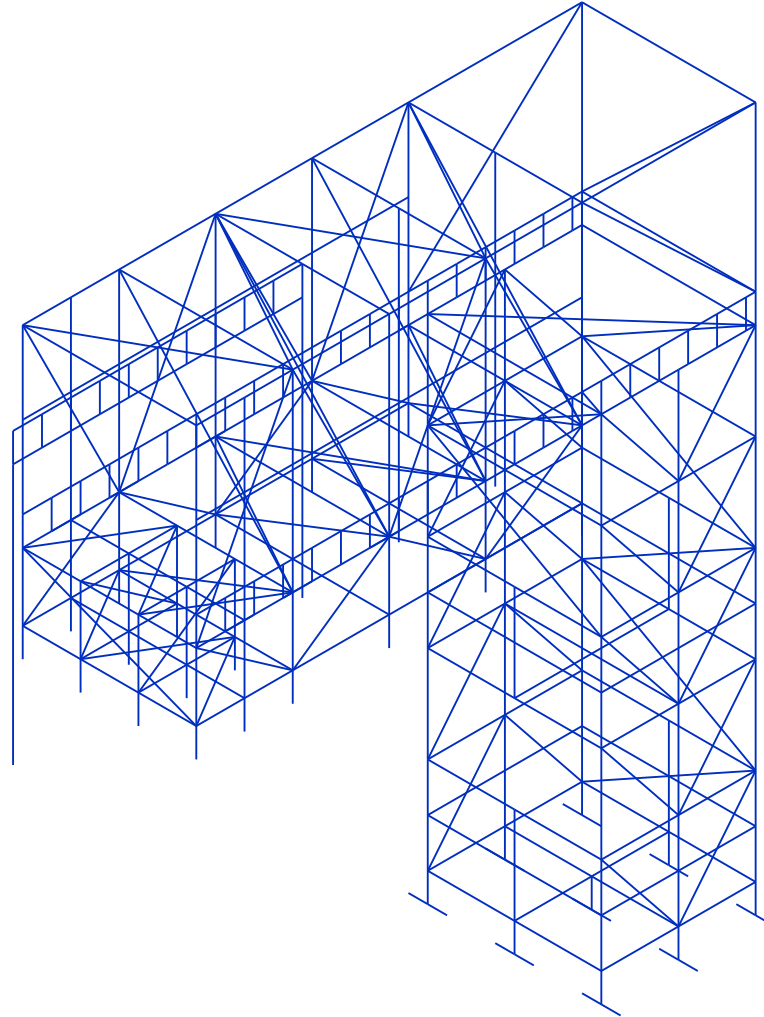
SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-040-SIDE-VIEW			 NLNG		
DRAWING TITLE PMS-WA-ARCO-040 - SIDE VIEW		SCALE NTS	DATE 29-JUN-2026	REV 0			
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
						SHEET 2 OF 4	SIZE A4



PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-040-PLAN-VIEW			 NLNG	
DRAWING TITLE PMS-WA-ARCO-040 - PLAN VIEW		SCALE NTS	DATE 29-JUN-2026	REV 0		
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	COMPANY	
					SHEET 3 OF 4	SIZE A4



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-040-3D-VIEW			 NLNG	
DRAWING TITLE PMS-WA-ARCO-040 - 3D VIEW		SCALE NTS	DATE 29-JUN-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
					SHEET 4 OF 4	SIZE A4

3D Isometric View

